Reliable Team Assistant: Phase 1

Yiting Dai | Solo | CSE 291 A Systems, Option 5 | October 24, 2026

Question and implemented scope

Can dependency-aware recovery preserve evidence-backed diagnosis quality with less recomputation after a trusted invalidation event? Configuration and telemetry roles collect read-only snapshots; a coordinator diagnoses. The runtime records consumed versions, invalidates dependent work, retains independent artifacts, rejects obsolete publications and links corrections. Missing recorded provenance triggers restart in development controls; this fallback was not exercised in the formal batch. Arbitrary error discovery, operational writes and invisible dependencies are outside the implemented guarantee.

Workload and frozen protocol

Cloud-OpsBench is pinned to ea05dafbc87b5985df8721622d63d4229c2b4224. Formal cases boutique/runtime/32, service/12 and performance/5 each run twice, reversing restart/selective order on repeat 2. These are three independent cases, not six. TritonAI api-glm-5.3 uses low reasoning, temperature 0 and 8192 output tokens. Course DataHub CPU 2/8 GB runs isolated Python 3.11.9 with exact dependency pins. Source 1380c6b identifies execution; the complete freeze is archived.

The user advanced launch to October 24 before formal observations were inspected. Cases, order, model, budgets and scoring were unchanged. Course acceptance of the explicitly user-selected benchmark remains unverified. Earlier development experiments and the 17-call TritonAI seen-case workflow remain separate feasibility evidence. No email or assignment submission was sent.

Formal quality results

All 6 units completed. Strict pass counts over all 6 assigned units: clean 4/6; restart 3/6; selective 3/6; prevention 3/6; expiry-restart 2/6; expiry-selective 2/6; no-recovery 0/6. For numeric-port recovery, only 4 units were eligible: restart and selective each passed 3/4, with different failed units. The other 2 units remain not-injected, not quietly excluded. Collection is not a diagnosis endpoint. No-recovery fails the current-publication criterion by construction; 0/6 does not imply six incorrect diagnoses. Prevention checks source integrity, not arbitrary semantic truth.

Strict success requires diagnosed status, native joint root/target accuracy and evidence closure, input-supported reasoning, current publication, valid output contract and budget. Input-only judgments on all 36 produced answers were hash-sealed before native labels were accessed. Codex performed the review; it is not independent human validation. All 54 assigned arm slots remain in scoring. Provisional publication does not mean operational approval.

Costs, limits and next steps

What the experiment establishes

Selective recovery used 1 new call versus 3 for restart in each of 4 eligible units: 33,260 versus 64,902 aggregate incremental tokens, 48.8% fewer. This excludes shared-prefix costs and is not end-to-end service savings. Equal 3/4 quality pass counts do not prove per-case equivalence or statistical non-inferiority. When all work expired, both methods needed 3 calls on all 6 units, demonstrating the conservative restart boundary.

No corrupted port was adopted in the four corrupted answers. This batch therefore supports invalidation/recomputation behavior and cost bounds, not correction of a propagated wrong diagnosis. Two service units had no eligible returned YAML field; no retargeting or rerun was performed. Service answers lacked discriminating evidence; other failures included wrong root labels, malformed quotations and insufficient distinct citations. Completion is not quality success.

Budget reconciliation

The batch finished with exit 0, 92 calls, 519,696 tokens and 0 API errors. Response-header costs total $0.40225095, matching refreshed account spend $0.62669165 minus $0.22444070 before the run. The immediate after-run balance lagged; that original receipt remains preserved. Conservative accounting is $0.66398535, below $20;92 calls are below 102. Remaining course balance was $49.37330835. No personal billing or additional inference for scoring.

Plan after this immutable evaluation

First independently audit evidence support, freshness and reproducibility. Preserve this run and all negative outcomes. Any improved query policy, semantic fault design or broader workload requires a new development/protocol version and new independent evaluation cases; these exposed cases must not be reused as unseen tests. Multi-group scheduling is the first extension and gated procedural learning the second, after the core claims are documented. Larger independent samples are needed for generalization.

Artifact and disclosure

The evidence ZIP includes raw requests/responses, ledgers, faults, frozen configuration, 36-row input review/seal, 54-row native/joint scores, cost summary, code and dependency lock. Static snapshots and constructed invalidation are not live production histories; the model alias is mutable. AI use: Codex assisted implementation, experiment tooling, analysis and report drafting. Student review is required before submission. Sources: github.com/LLM4Ops/Cloud-OpsBench; course Option 5 overview; Piazza 46; docs.tritonai.ucsd.edu/models.

Phase 1: Reproduction and Timeline Appendix

Reproducible artifact

The accompanying ZIP includes pinned native evaluator source, MIT license, three scored label sets, code, tests and a portable offline verifier. README-REPRODUCTION.txt gives clean extraction commands and the pinned source acquisition command. Original formal review judgments and their seal are unchanged. The verifier resolves the review by sealed content hash, not the author-specific absolute path.

Measurement boundaries

The stored seconds field sums API-call durations; integrated end-to-end latency and formal tracking overhead were not recorded. No numeric replacement is invented. Four eligible injections did not cause adoption of the false port; two units were ineligible. Equal 3/4 recovery success counts have different failed cases and do not establish quality equivalence.

Separate causal-summary development

On seen runtime/1, replacing the summary protocol httpGet with grpc while preserving raw evidence did not propagate (0/1): the coordinator rejected it. Restart/selective regenerated the correct summary, not a wrong final diagnosis. They used 4/2 calls, 18,392/11,959 incremental tokens and 10.250/4.759 seconds measured recovery wall time. This is a single-run observation, not generalization. The separate batch used 11 calls, 57,541 tokens and $0.04764120; full traces are in development/causal-summary. The formal seal is unchanged.

Completed October 24

Formal scoring, billing reconciliation and immutable evidence preservation. Portable saved-output verification is separate from fresh inference and cannot establish repeatability of a mutable model alias. Input-only judgments were produced by Codex, not independent human review.

October 24-November 4

Finish clean extraction checks and causal-error development on seen cases; keep failures and ignored injections. Prepare the Phase 1 submission by the updated November 4 course target; confirm exact current time and upload slots before submission. Nothing in this release has been submitted.

Next experimental milestone

Freeze a new causal fault protocol only after development establishes propagation. Compare no recovery, restart and selective recovery from the same failure state. Include shared/complete invalidation and missing provenance. Newly selected evaluation cases must be separate from all exposed cases.

Remaining term

Analyze paired quality before cost; add scheduling or learned-memory extensions only after the core recovery claim is supported. Recheck API expiry and budget before any additional online batch. Final artifact/report preparation targets late November to December; verify official deadlines. Benchmark acceptance remains unverified.